Autonomous Physics-Based Modeling of Retention Loss in 3-D NAND Flash Memory in Cryogenic condition

EDL Manuscript Change & Progress

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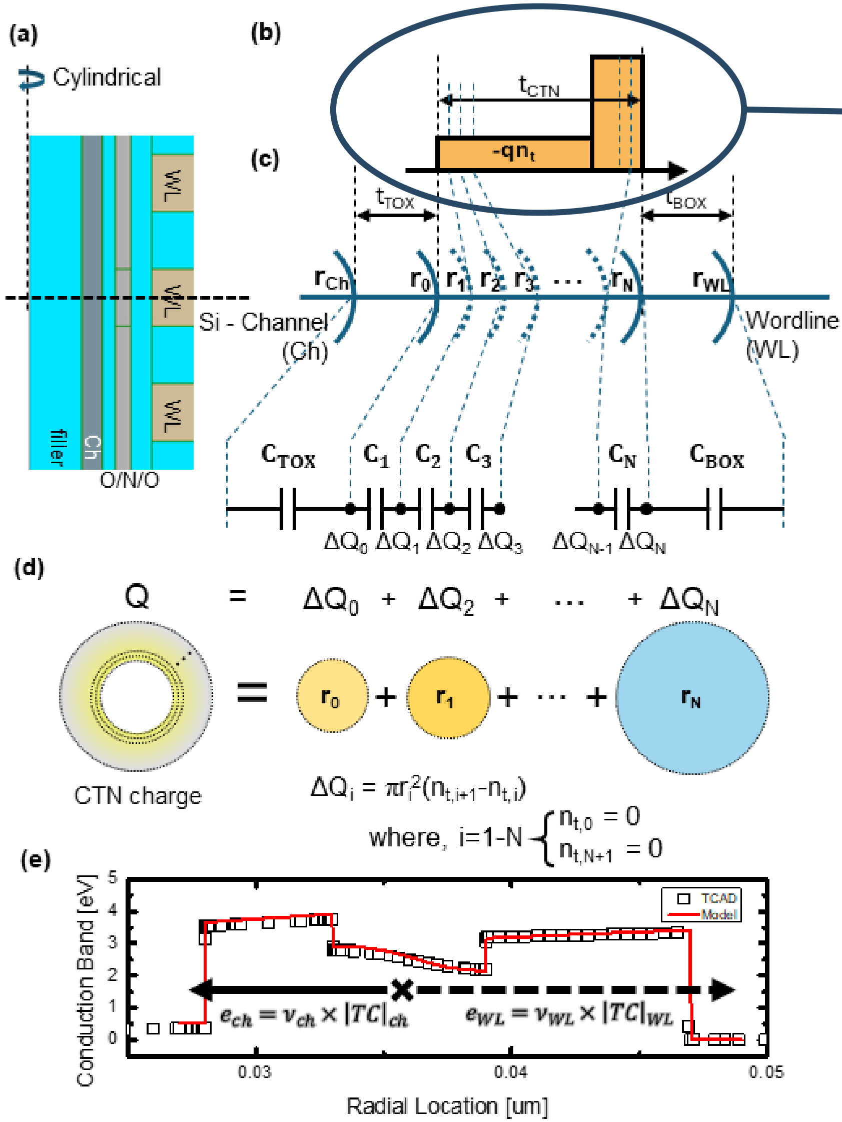
Detailed Poisson Solution

Fig. 1



$$n_c(r) = \frac{J_{in}}{q\mu F_{TOX-CTN}} \times \exp\left(-\int_{r_{TOX}}^{r}\left(\frac{\sigma_n v_{th} N_t^{av}}{\mu F_{CTN}} + \frac{1}{r'} + \frac{dF_{CTN}}{F_{CTN} dr'}\right)dr'\right).$$

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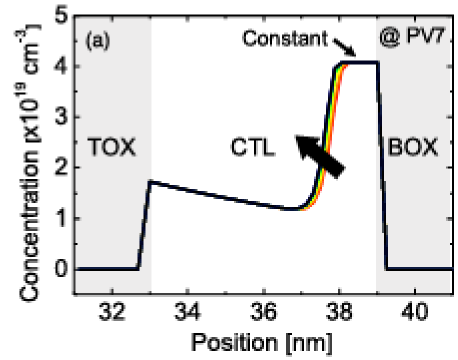


JJAP 65 (2026) 011004

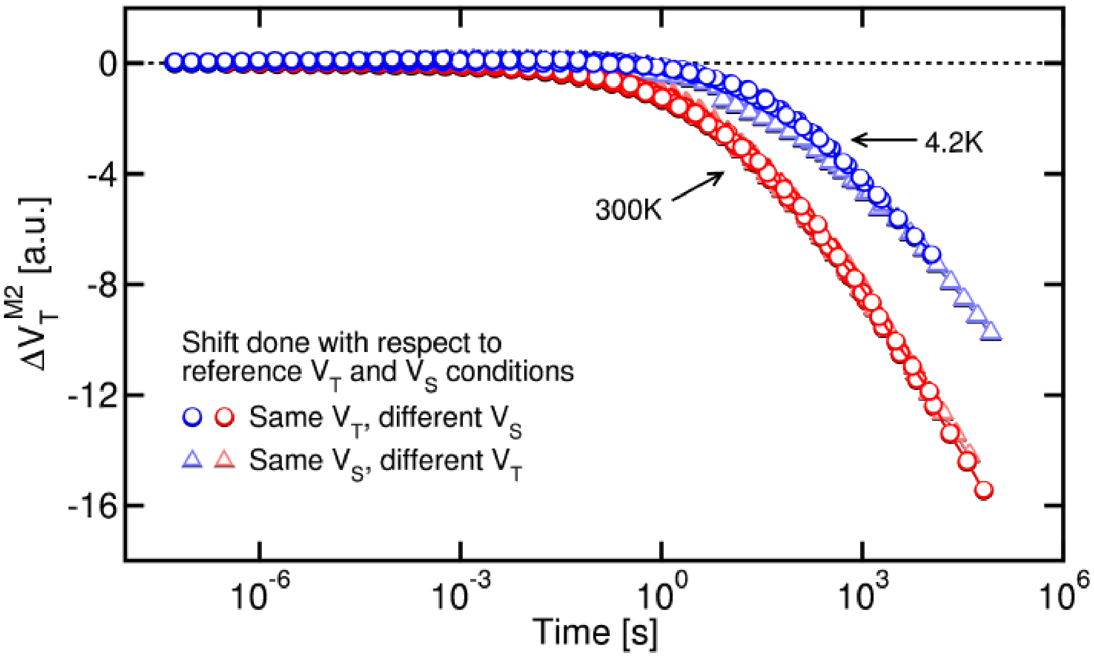
(b) pgm charge

Charge distribution model adapted

- ➢ 최해찬/유진일 TED 2024
- ➢ 이찬 JJAP 2026

Cryogenic and Detrapping

Measurement (Refaldi, IRPS 2025)



- Cryogenic condition (~ 4.2K) : physical separation of detrapping effect

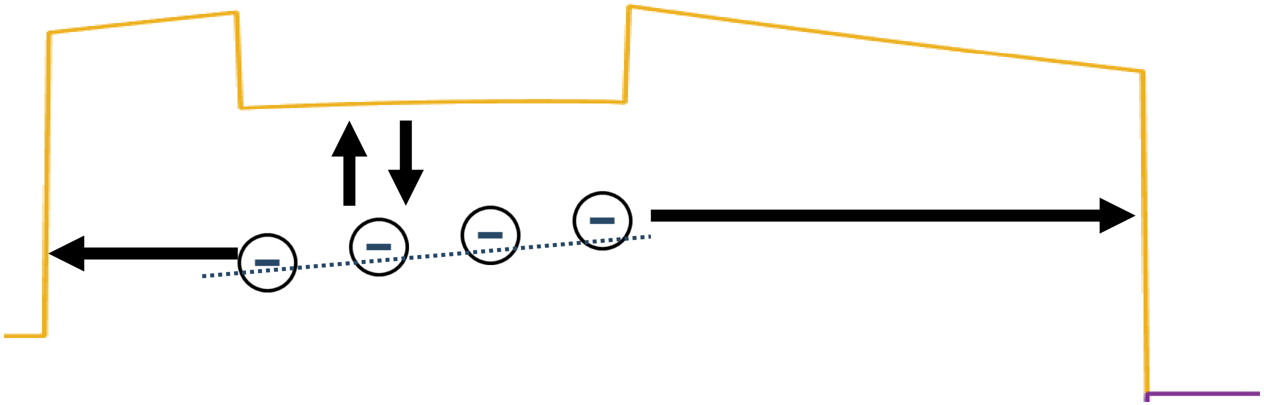
- Separation of detrapping effect measurement

****Time is in not in A.U. : calibration of time constant can be done**

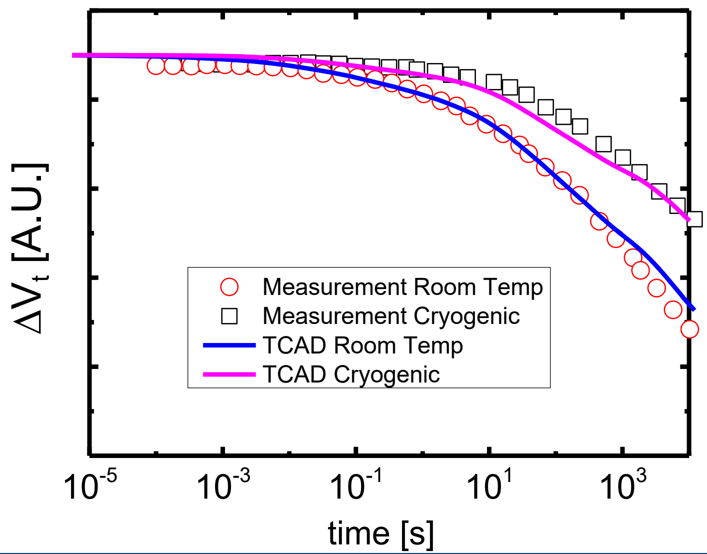
Cryogenic and Detrapping

Deactivate Conduction band – trap interaction

- Cryogenic
- $\sigma \to 0$



Measurement calibration result:



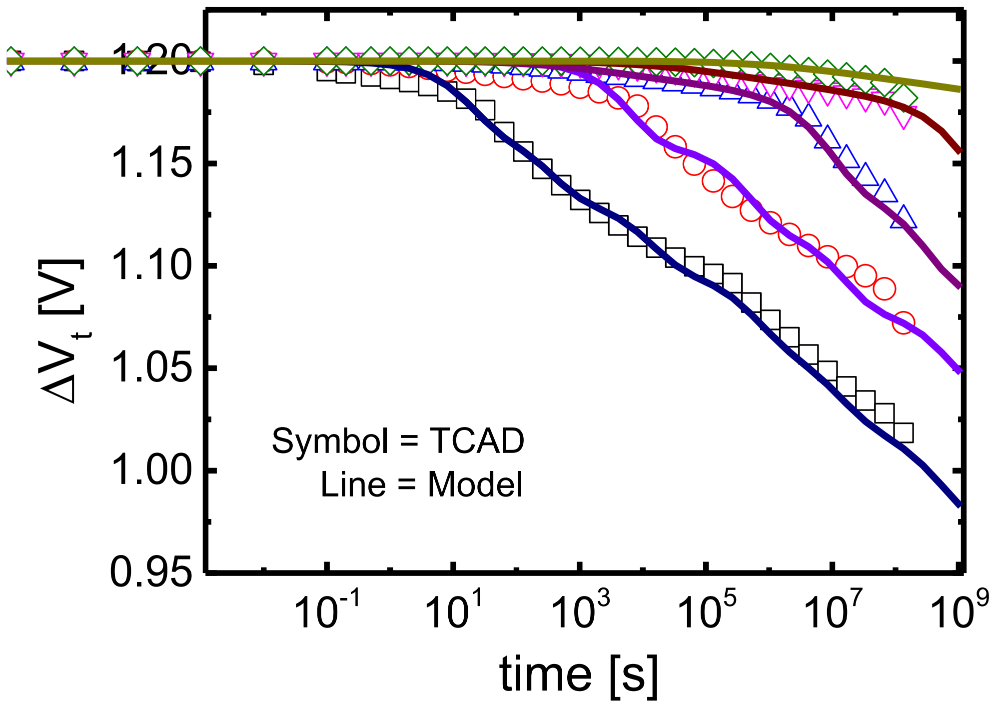
Used scale

O/N/O = **3**/6/8 [nm]

(cf. DRL spec: 5/6/8 [nm])

Autonomous Modeling & Result

TCAD – model Matching



TOX variation
3/3.5/4/4.5/5 [nm]

Well match with
TCAD - model

Progress with paper

Done progress

- Measurement calibration figure added
- TCAD-model matching figure added

On-going progress

- Benchmark information
- Referencing